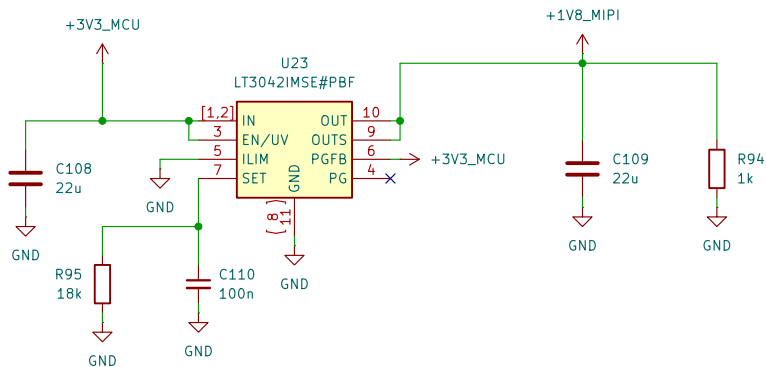
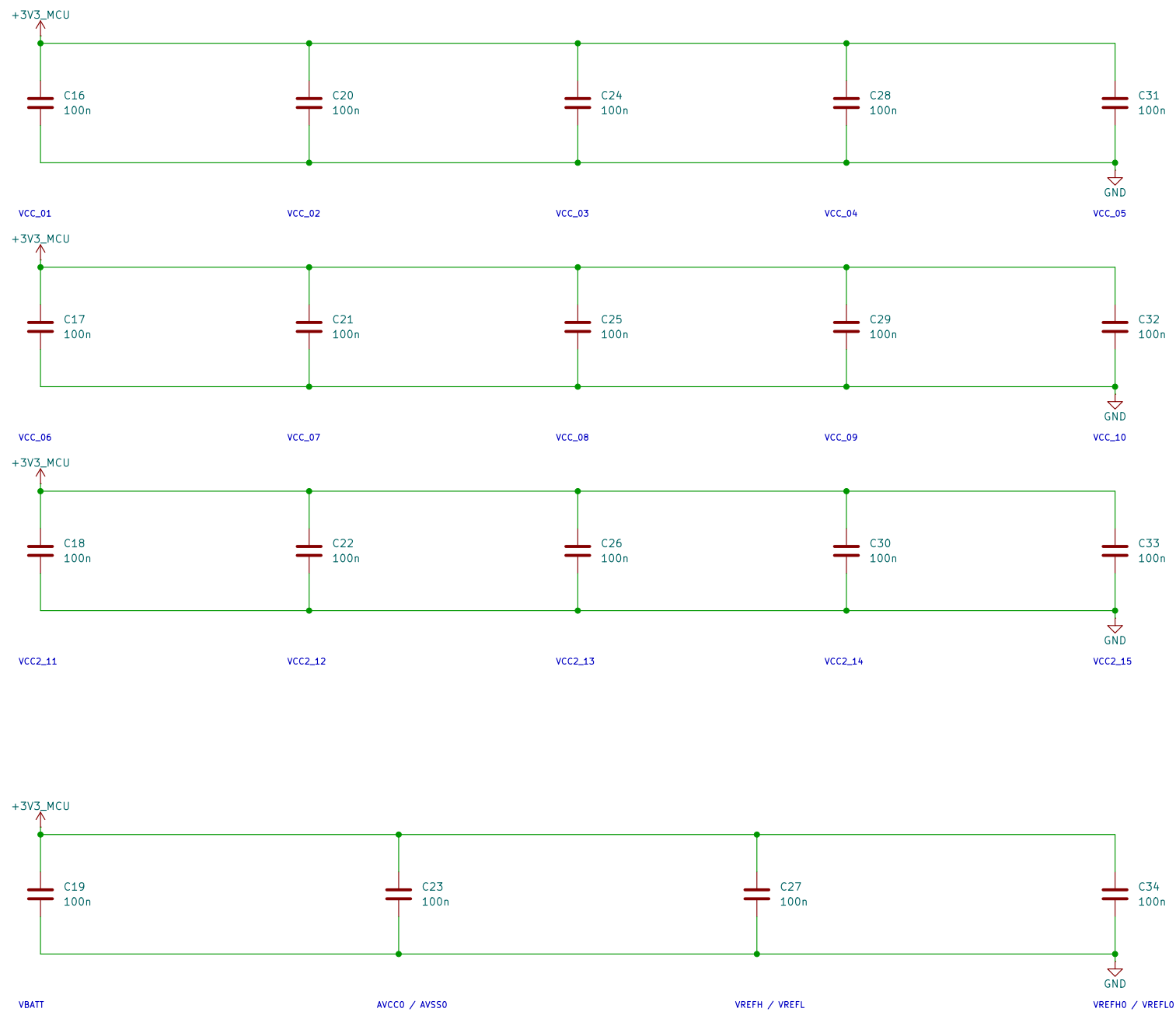


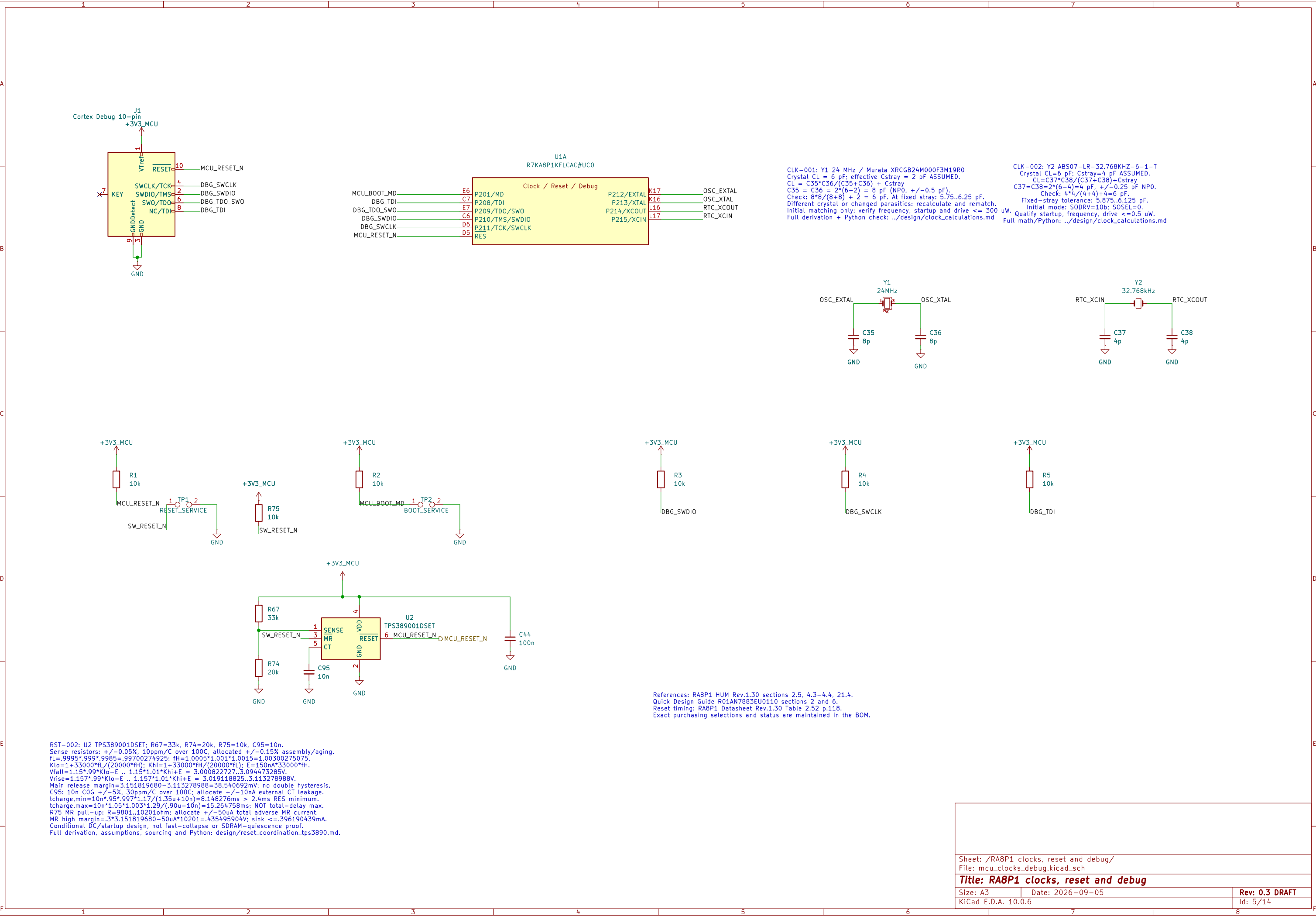


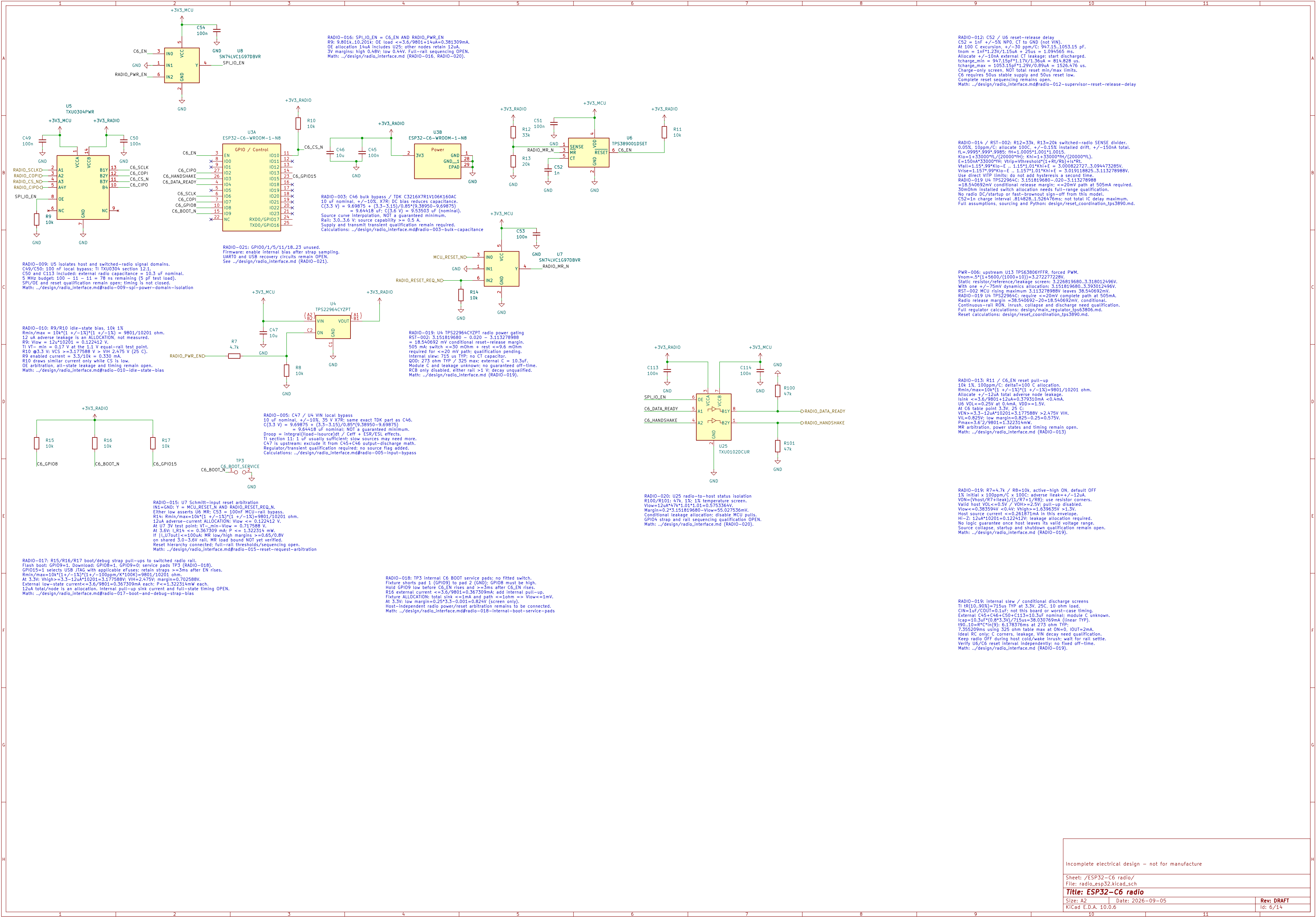
I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR

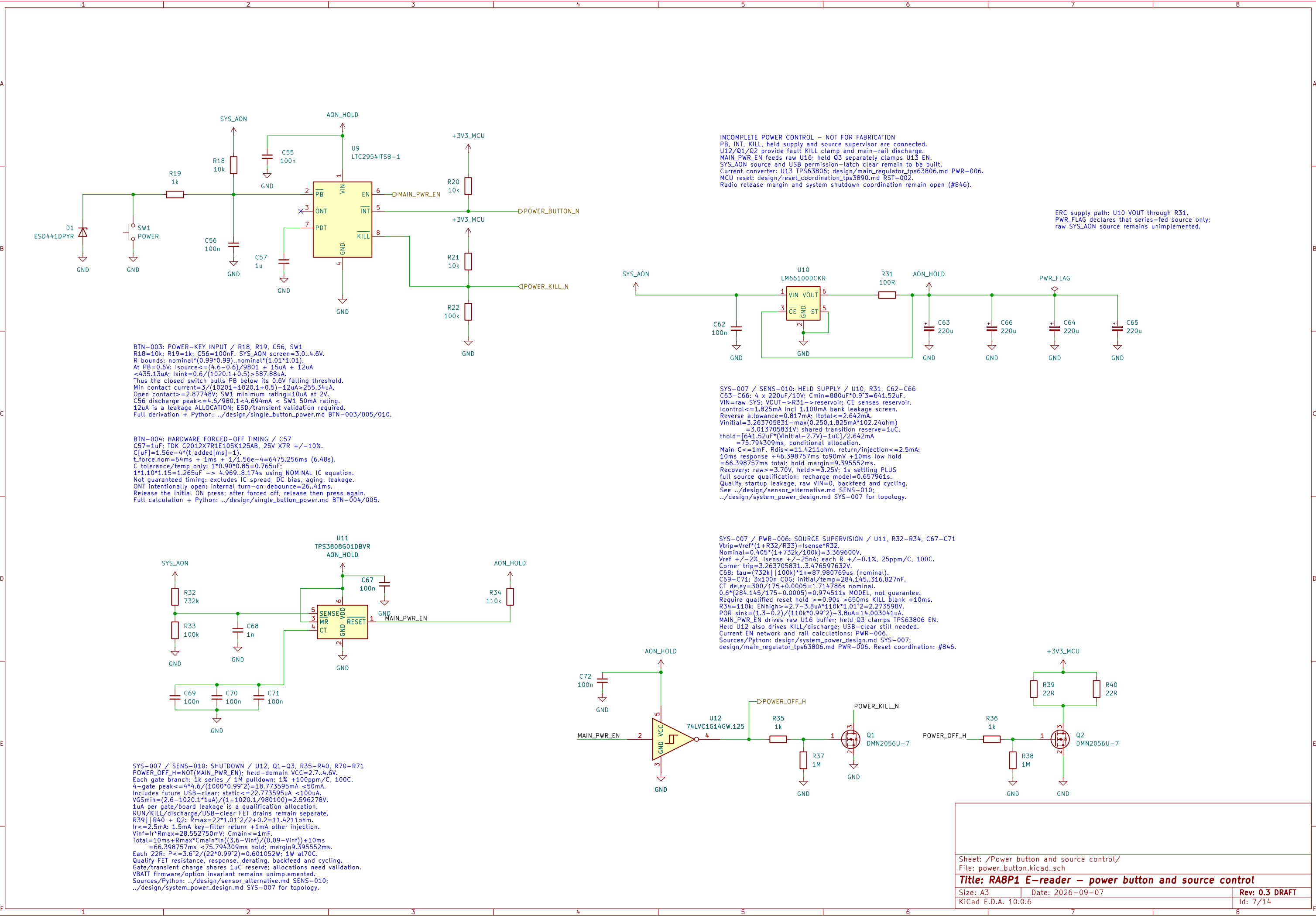


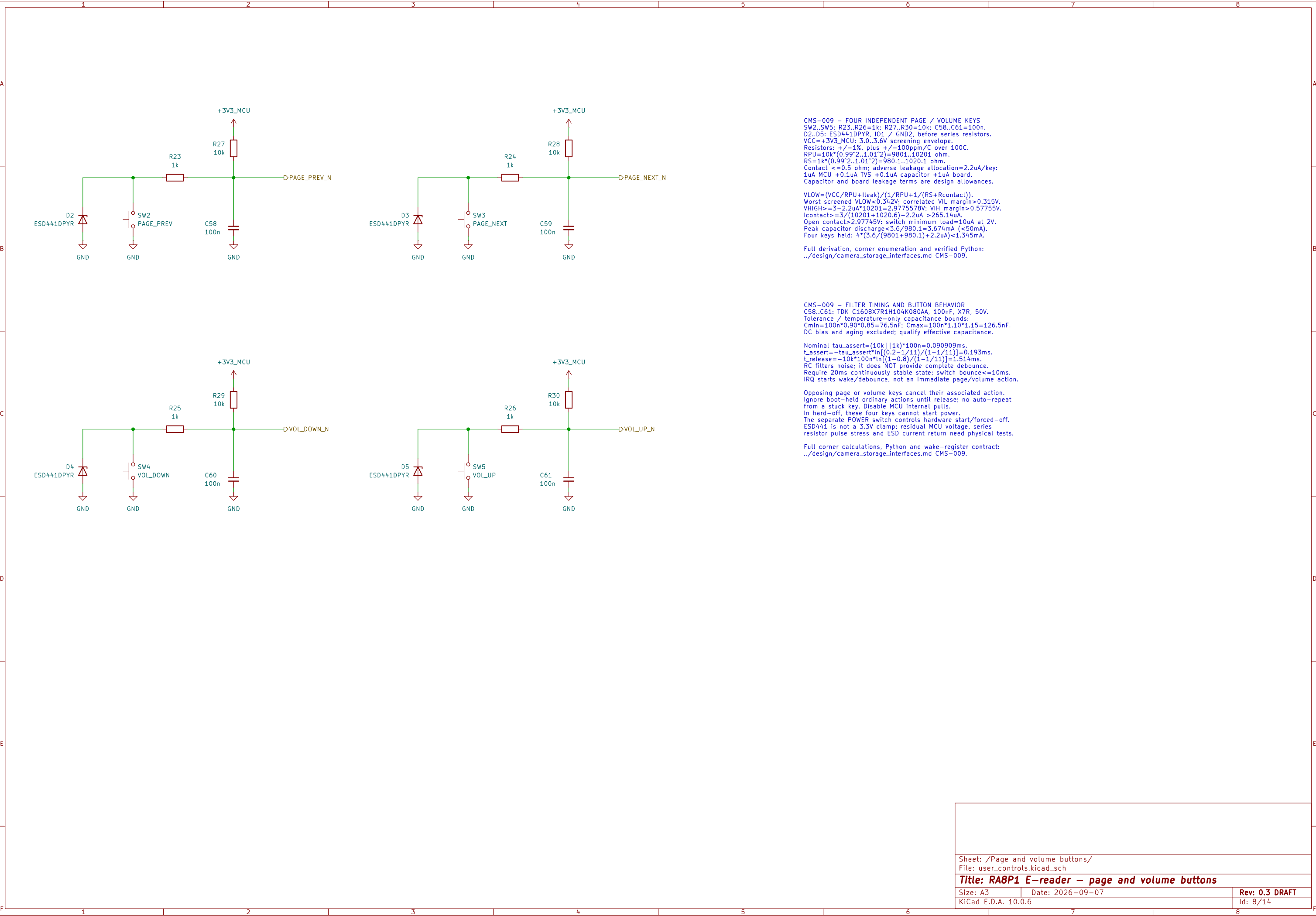
CMS-013; +20mA steady CSI branch planning allocation
4.1mA PHY + 1.989593mA bleed + 13mA allocated GND
+ 0.102mA SET + 0.100mA auxiliary = 19.291593mA.
13mA is an engineering envelope, not an interpolated guarantee.
Auxiliary allowance requires verification; startup/inrush excluded.
AVCC/MPI remains within existing MCU allocation.
Main subtotal: 2.075 + 0.020 = 2.095A; full system budget pending.
At 3.3V: 2.095A @ 3.3V = 6.9145W. Efficiency:
Input: 2.961817158A; converter loss = 3.369453727W
Source, current and thermal acceptance remain open.
Sources, conditions and Python: design/camera_storage_interfaces.md

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CMS-009 – FOUR INDEPENDENT PAGE / VOLUME KEYS
SW2..SW5; R23..R26=1k; R27..R30=10k; C58..C61=100n.
D2..D5: ESD441DPYR, IO1 / GND2, before series resistors.
VCC=+3V3_MCU; 3.0..3.6V screening envelope.
Resistors: +/-1%, plus +/-100ppm/C over 100C.
RPU=10k*(0.99^2..1.01^2)=9801..10201 ohm.
RS=1k*(0.99^2..1.01^2)=980.1..1020.1 ohm.
Contact <=0.5 ohm; adverse leakage allocation=2.2uA/key:
1uA MCU +0.1uA TVS +0.1uA capacitor +1uA board.
Capacitor and board leakage terms are design allowances.

$VLOW=(VCC/RPU+Ileak)/(1/RPU+1/(RS+Rcontact))$.
Worst screened VLOW<0.342V; correlated VIL margin>0.315V.
VHIGH>=3-2.2uA*10201=2.9775578V; VIH margin>0.57755V.
Icontact>=3/(10201+1020.6)-2.2uA >265.14uA.
Open contact>2.97745V; switch minimum load=10uA at 2V.
Peak capacitor discharge<3.6/980.1=3.674mA (<50mA).
Four keys held: 4*(3.6/(9801+980.1)+2.2uA)<1.345mA.

Full derivation, corner enumeration and verified Python:
../design/camera_storage_interfaces.md CMS-009.

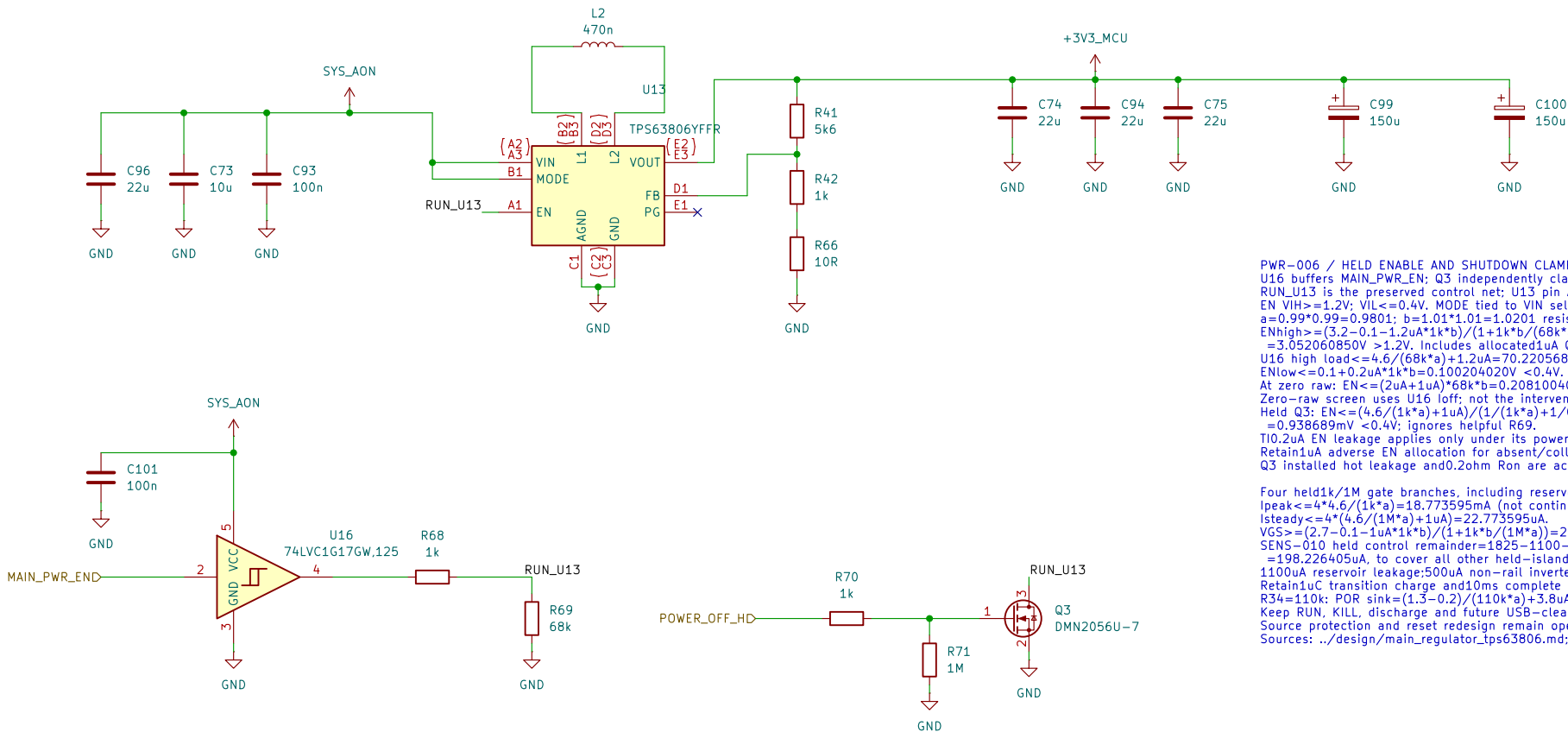
CMS-009 – FILTER TIMING AND BUTTON BEHAVIOR
C58..C61: TDK C1608X7R1H104K080AA, 100nF, X7R, 50V.
Tolerance / temperature-only capacitance bounds:
Cmin=100n*0.90*0.85=76.5nF; Cmax=100n*1.10*1.15=126.5nF.
DC bias and aging excluded; qualify effective capacitance.

Nominal tau_assert=(10k||1k)*100n=0.090909ms.
t_assert=-tau_assert*ln[(0.2-1/11)/(1-1/11)]=0.193ms.
t_release=-10k*100n*ln[(1-0.8)/(1-1/11)]=1.514ms.
RC filters noise; it does NOT provide complete debounce.
Require 20ms continuously stable state; switch bounce<=10ms.
IRQ starts wake/debounce, not an immediate page/volume action.

Opposing page or volume keys cancel their associated action.
Ignore boot-held ordinary actions until release; no auto-repeat
from a stuck key. Disable MCU internal pulls.
In hard-off, these four keys cannot start power.
The separate POWER switch controls hardware start/forced-off.
ESD441 is not a 3.3V clamp; residual MCU voltage, series
resistor pulse stress and ESD current return need physical tests.

Full corner calculations, Python and wake-register contract:
../design/camera_storage_interfaces.md CMS-009.

MAIN DIGITAL SUPPLY: TPS63806 / PWR-006
3.272277 V nominal; forced PWM; held shutdown clamp.
Divider and bypass calculations: design/main_regulator_tps63806.md
Reset coordination (#846), source protection and qualification remain open.
DEVELOPMENT SCHEMATIC – NOT FOR POWER-UP OR FABRICATION.



PWR-006 / FEEDBACK AND LOCAL PASSIVES
 $V_{nom} = 0.500 * (1 + 5600 / (1000 + 10)) = 3.272277228 \text{ V}$.
R41: 0.02%, 5ppm/C; R42: 0.01%, 2ppm/C; R66: 1%, 200ppm/C.
Installed bounds: initial*TCR(100C)*aging*assembly, then ohms.
 $R41 = 5600 * (1 + / - 0.0002) * (1 + / - 0.0005) * (1 + / - 0.001) * (1 + / - 0.0005) + / - 0.02 = 5587.669237..5612.349243 \text{ ohm}$.
 $R42 = 1000 * (1 + / - 0.0001) * (1 + / - 0.0002) * (1 + / - 0.001) * (1 + / - 0.0005) + / - 0.01 = 998.190970..1001.810970 \text{ ohm}$.
 $R66 = 10 * (1 + / - 0.01) * (1 + / - 0.02) * (1 + / - 0.01) + / - 0.05 = 9.554980..10.455020 \text{ ohm}$; one combined aging/assembly 1%.
 $V_{lo} = 0.495 * (1 + R41_{min} / (R42_{max} + R66_{max})) - 100\text{nA} * R41_{max}$.
 $V_{hi} = 0.505 * (1 + R41_{max} / (R42_{min} + R66_{min})) + 100\text{nA} * R41_{max}$.
 $V_{static} = 3.226819680..3.318012496 \text{ V}$ (FPWM only).
One combined +/-75mV regulation/ripple/transient/routing allowance:
 $V_{complete} = 3.151819680..3.393012496 \text{ V}$; qualification required.
SDRAM static VOH margin = $2.4 - 0.7 * 3.393012496 = 24.891253\text{mV}$.
This is not measured SI/noise headroom or reset qualification.

L2: 470nH +/-20%; catalog spread alone is insufficient.
Screen incoming 0.412..0.518uH; allocated operating +/-10%
=>0.3708..0.5698uH versus TI0.37..0.57uH. Qualify hot/current.
C74/C75/C94: $3 * 22\text{uF} * 0.8 * 0.85 * 0.6 = 26.928\text{uF} > \text{TI}21\text{uF}$.
C96: $22\text{uF} * 0.8 * 0.85 * 0.6 = 8.976\text{uF} > \text{TI}4\text{uF}$ input minimum.
0.6 is allocated combined bias/aging retention, NOT guaranteed.
Samsung typical output-bias retention at 3.393012496V = 90.221554%.
Input bias, aging, mounted impedance and startup need verification.
Full assumptions, sources and executable Python: click this note.

PWR-006 / HELD ENABLE AND SHUTDOWN CLAMP
U16 buffers MAIN_PWR_EN; Q3 independently clamps EN during raw collapse.
RUN_U13 is the preserved control net; U13 pin A1 is now EN.
EN VIH>=1.2V; VIL<=0.4V. MODE tied to VIN selects forced PWM.
 $a = 0.99 * 0.99 = 0.9801$; $b = 1.01 * 1.01 = 1.0201$ resistor factors.
 $\text{EN}_{high} > = (3.2 - 0.1 - 1.2\text{uA} * 1\text{k} * b) / (1 + 1\text{k} * b / (68\text{k} * a))$
 $= 3.052060850\text{V} > 1.2\text{V}$. Includes allocated 1uA Q3 off leakage.
U16 high load $< = 4.6 / (68\text{k} * a) + 1.2\text{uA} = 70.220568\text{uA} < 100\text{uA}$.
 $\text{EN}_{low} < = 0.1 + 0.2\text{uA} * 1\text{k} * b = 0.100204020\text{V} < 0.4\text{V}$.
At zero raw: $\text{EN} < = (2\text{uA} + 1\text{uA}) * 68\text{k} * b = 0.208100400\text{V}$.
Zero-raw screen uses U16 loff; not the intervening undefined region.
Held Q3: $\text{EN} < = (4.6 / (1\text{k} * a) + 1\text{uA}) / (1 / (1\text{k} * a) + 1 / 0.2\text{ohm})$
 $= 0.938689\text{mV} < 0.4\text{V}$; ignores helpful R69.
TI0.2uA EN leakage applies only under its powered test conditions.
Retain 1uA adverse EN allocation for absent/collapsing raw.
Q3 installed hot leakage and 0.2ohm Ron are acceptance allocations.

Four held 1k/1M gate branches, including reserved USB-clear:
 $I_{peak} < = 4 * 4.6 / (1\text{k} * a) = 18.773595\text{mA}$ (not continuous rating).
 $I_{steady} < = 4 * (4.6 / (1\text{M} * a) + 1\text{uA}) = 22.773595\text{uA}$.
 $V_{GS} > = (2.7 - 0.1 - 1\text{uA} * 1\text{k} * b) / (1 + 1\text{k} * b / (1\text{M} * a)) = 2.596278\text{V}$.
SENS-010 held control remainder = $1825 - 1100 - 500 - 4 - 22.773595$
 $= 198.226405\text{uA}$, to cover all other held-island loads.
1100uA reservoir leakage; 500uA non-rail inverter-input allocation.
Retain 1uC transition charge and 10ms complete response allocation.
R34 = 110k; POR sink = $(1.3 - 0.2) / (110\text{k} * a) + 3.8\text{uA} = 14.003041\text{uA}$.
Keep RUN, KILL, discharge and future USB-clear drains separate.
Source protection and reset redesign remain open; no power-up approval.
Sources: ../design/main_regulator_tps63806.md; SENS-010 revision.

PWR-006 / LOAD, STORAGE AND SHUTDOWN
Allocation: 1.8A operating; 2.25A cold-start screen, not measured load.
 $P_{out} = 3.393012496 * 1.8 = 6.107422493\text{W}$.
At allocated 75% efficiency: $P_{in} = 8.143229991\text{W}$; $I_{in@3.2V} = 2.544759372\text{A}$.
Loss = 2.035807498W; NOT an acceptable continuous thermal result.
At 90% efficiency: loss = 0.678602499W.
TJ screen = $70\text{C} + 78.8\text{C/W} * 0.678602499\text{W} = 123.474\text{C}$.
TI board theta-JA is not the final PCB thermal resistance.
2.25A screen: $P_{in} = 3.393012496 * 2.25 / 0.75 = 10.179037489\text{W}$.
Converter limit, source limit, startup and thermal envelope need validation.

PWR-006 storage checkpoint (nominal, not effective minimum):
Direct main 469.21uF + C43 through FB1 10uF + key filters 0.4uF
=>479.61uF cold; radio-on adds 10.2uF =>489.81uF.
C99/C100 remain 150uF each, C94 adds 22uF to the output.
Old bootstrap/private VCC/VC capacitors were not main-rail storage.
Whole-main ceiling remains 1mF; reconcile later peripheral additions.
Cout lower acceptance and distributed PDN/startup are separate checks.

SENS-010 discharge revision: $R_{max} 11.4211\text{ohm}$; return/injection $< = 2.5\text{mA}$.
Includes 1.5mA key return + 1mA ether backfeed: $V_{inf} = 0.028552750\text{V}$.
 $t = 1\text{m} * 11.4211\text{ohm} * \ln((3.6 - V_{inf}) / (0.09 - V_{inf})) + 10\text{ms} + 10\text{ms}$
 $= 66.398757\text{ms}$ total, below held 75.794309ms screen.
Includes 10ms command/clamp/discharge response and 10ms below 90mV.
Held budget: 1.825mA control + 0.817mA reverse; 1uC transition reserve.
198.226405uA control remainder must cover other held-island loads.
SYS_AON source protection, loaded startup and fast faults remain open.
RST-002 / RADIO-019: release margin remains conditional on rail/path bounds.
Do not power up until reset coordination and source protection are complete.
Full assumptions: ../design/main_regulator_tps63806.md;
shutdown revision/Python: ../design/sensor_alternative.md SENS-010.

Sheet: /Main 3V3 digital supply/
File: main_3v3_supply.kicad_sch

Title: RA8P1 E-reader – main digital 3.3 V supply

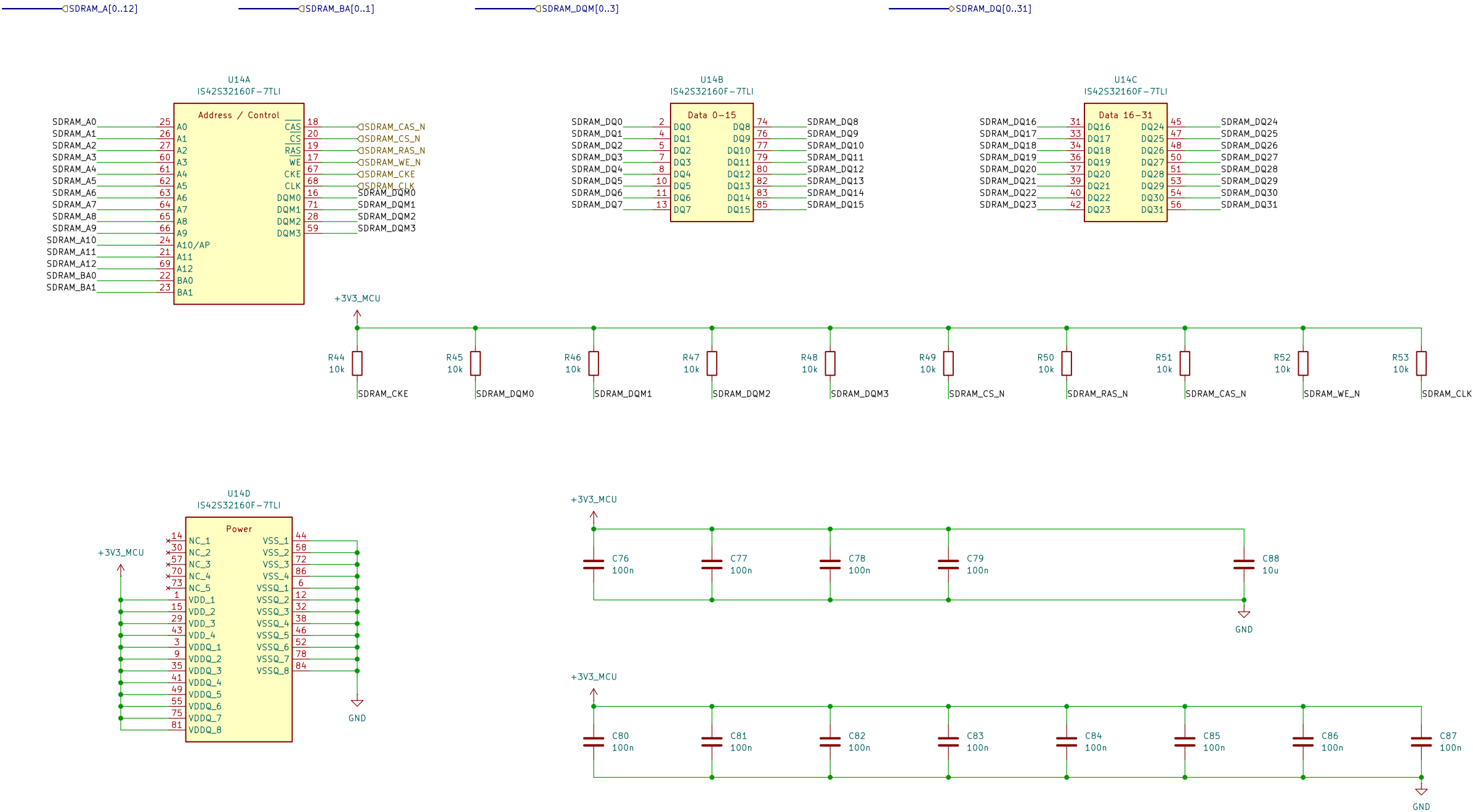
Size: A3 Date: 2026-09-08

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.6

Id: 9/14

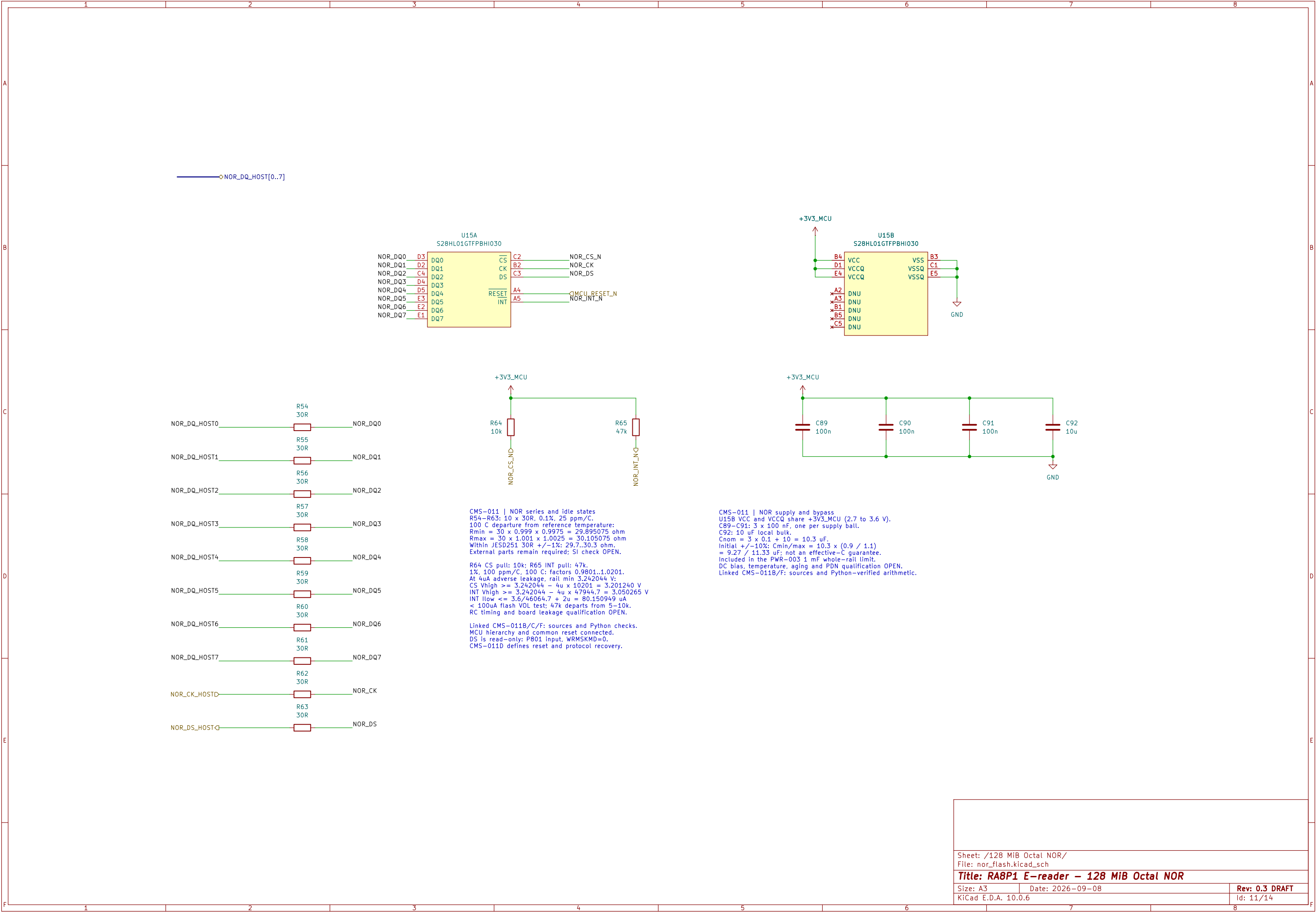
64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.

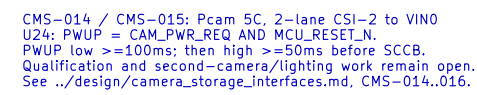


CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT| <=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

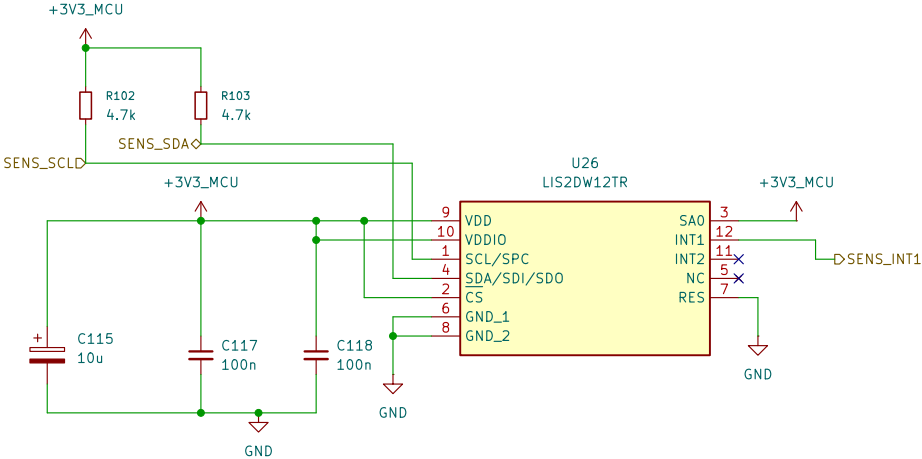
CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail:
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.





Sheet: /Pcam 5C camera/ File: camera.kicad_sch		
Title:		
Size: A3	Date:	Rev:
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ORIENTATION SENSOR – WIP
U26: LIS2DW12TR; ST DS11811 Rev 9 pin map.
VDD pin 9 / VDDIO pin 10: +3V3_MCU; CS / SA0 high for I2C address 0x19.
C117/C118: local 100n bypass; GND pins 6/8 and RES pin 7 grounded.
C115: 10u/16V aluminum EEE–FM1C100R; place near VDD pin 9 with C117.
R102/R103: 4.7k SCL/SDA pull–ups; obsolete ADXL VREG network removed.
Host connected: SCL P410/K13; SDA P409/R16; INT1 P306/D11.
Wait 20ms after supply valid. Power–cycle below 100mV for >=10ms.
Pending: sourcing fields, pin modes, timing/power/passive qualification.
See design/sensor_alternative.md; circuit qualification remains open.



Sheet: /Orientation sensor/
File: orientation_sensor.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.6

Rev:

Id: 14/14